

1 Methods of proof

In the domain of mathematics, establishing the truth of a proposition involves deriving it from a set of assumed true mathematical statements known as hypotheses. This deductive process relies on logical principles, incorporating elements of natural language, and making use of mathematical constructs such as axioms, theorems, propositions, lemmas, and corollaries. It is noteworthy that while hypotheses are typically explicitly stated in the proposition under consideration, the same clarity may not extend to the axioms, theorems, propositions, lemmas, and corollaries utilized in the course of the proof. Generally, the proof of a proposition Q involves demonstrating an implication $P \Rightarrow Q$, where P represents the hypothesis, and the objective is to establish the truth of Q . To obtain a proof, we must:

- respect the rules of logic,
- know the different types of proofs, and
- exercise good sense and rigor.

In this section, we outline the main strategies for proving a mathematical statement.

1.1 Proof by deduction (Direct proof)

Deductive reasoning is the most common type of proof. Starting with a hypothesis, a logical line of reasoning is constructed, leading to a conclusion. This method is based on the tautology

$$P \wedge (P \Rightarrow Q) \Rightarrow Q.$$

The proposition P represents the set of hypotheses and Q the conclusion to be reached, given that the implication $P \Rightarrow Q$ is true.

Example 1

1. *Let us show that if a and b are odd integers, then $a + b$ is an even integer. Assume that a and b are odd integers. Then, there are integers k and k' such that $a = 2k + 1$ and $b = 2k' + 1$. Then, we have $a + b = 2k + 1 + 2k' + 1 = 2(k + k' + 1)$, which shows that $a + b$ is an even integer.*
2. *Let us show that any positive integer that is a square has an odd number of divisors. Let a be a perfect square integer. Then among the divisors of a , we have 1 and a itself. Since $a = b^2$, a admits b as a divisor. This already gives three divisors: 1, a , and b . Furthermore, for any other divisor c of a , different from 1, a , and b , there must be an integer d such that $a = cd$. The integer d is also a divisor of a and must be different from 1, a , b , and c . Therefore, if there are divisors other than 1, a , and b , they must come in pairs. The total number of divisors is therefore odd.*

1.2 Proof by contrapositive

Proof by contrapositive is based on the mathematical equivalence

$$(P \Rightarrow Q) \Leftrightarrow (\overline{Q} \Rightarrow \overline{P}).$$

In some cases, showing the implication $\overline{Q} \Rightarrow \overline{P}$ is simpler than showing $P \Rightarrow Q$. In a proof by contrapositive, you start by assuming that Q is false (i.e., $\overline{Q}$ is true). You then demonstrate a series of direct implications leading to the conclusion that P is also false (i.e., $\overline{P}$ is true). Consequently, if P being false implies Q is false, then $P \Rightarrow Q$ must hold true.

Example 2

1. Let $n \in \mathbb{N}$. We will show that if n^2 is even, then n is even. We will prove the contrapositive of this proposition. Assume that n is odd. Then there exists $k \in \mathbb{N}$ such that $n = 2k + 1$. Thus,

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Let $k' = 2k^2 + 2k$. Then $n^2 = 2k' + 1$, which is odd. Thus, if n is odd, then n^2 is odd. By the contrapositive, if n^2 is even, then n is even.

2. Let m and n be two nonzero natural numbers. We will show that if mn is odd, then both m and n must be odd.

Assume that m is even. Then there exists $k \in \mathbb{N}$ such that $m = 2k$. Therefore,

$$mn = (2k)n = 2(kn),$$

which is even. The case where n is even is similar: if n is even, then there exists $l \in \mathbb{N}$ such that $n = 2l$, and thus,

$$mn = m(2l) = 2(ml),$$

which is also even.

Thus, if either m or n is even, then mn is even. By the contrapositive, if mn is odd, then both m and n must be odd.

1.3 Proof by contradiction

This method is based on the mathematical equivalence

$$\overline{(P \Rightarrow Q)} \Leftrightarrow P \wedge \overline{Q}.$$

In this case, we assume Q to be false and P to be true. If in the course of reasoning we get a contradiction, $P \wedge \overline{Q}$ cannot be true, then

$$P \Rightarrow Q$$

is true, we deduce that Q is true.

Example 3

1. Let us show that $\sqrt{p} \notin \mathbb{Q}$ for any prime p . Assume for contradiction that $\sqrt{p} = \frac{p_1}{q_1}$, where $p_1, q_1 \in \mathbb{Z}^*$ (nonzero integers) and $\gcd(p_1, q_1) = 1$. Then,

$$\left(\frac{p_1}{q_1}\right)^2 = p.$$

This implies

$$\frac{p_1^2}{q_1^2} = p.$$

Rearranging gives

$$p_1^2 = pq_1^2. \quad (1)$$

Since p_1^2 is divisible by p , p_1 must be divisible by p . Let $p_1 = pr$ for some $r \in \mathbb{N}$. Substituting $p_1 = pr$ into the equation 1 gives

$$(pr)^2 = pq_1^2,$$

which simplifies to

$$p^2r^2 = pq_1^2.$$

Dividing both sides by p , we obtain

$$pr^2 = q_1^2.$$

This shows that q_1^2 is divisible by p , so q_1 must also be divisible by p . However, this contradicts our initial condition that p_1 and q_1 are coprime ($\gcd(p_1, q_1) = 1$). Hence, the assumption that $\sqrt{p}$ can be expressed as $\frac{p_1}{q_1}$ with $p_1, q_1 \in \mathbb{Z}^*$ is false. Therefore, $\sqrt{p} \notin \mathbb{Q}$ for any prime p .

2. Let us show that for any prime integer p and for any rational numbers x, y, z , and t , if

$$x + y\sqrt{p} = z + t\sqrt{p},$$

then

$$x = z \quad \text{and} \quad y = t.$$

Assume that

$$x + y\sqrt{p} = z + t\sqrt{p} \quad \text{and} \quad y \neq t.$$

We need to show that this assumption leads to a contradiction. Starting from the equality:

$$x + y\sqrt{p} = z + t\sqrt{p},$$

subtract $z + t\sqrt{p}$ from both sides gives

$$x + y\sqrt{p} - z - t\sqrt{p} = 0.$$

This simplifies to

$$(x - z) + (y - t)\sqrt{p} = 0.$$

Suppose $y \neq t$. Then

$$\frac{z - x}{y - t} = \sqrt{p} \in \mathbb{Q},$$

which is impossible from example 1). Thus, $y = t$ and so $x = z$.

3. Here is perhaps the oldest usage of reasoning by contradiction, attributed to Euclid, to demonstrate that the set of prime numbers is infinite. Suppose there is only a finite number of prime numbers, say $p_1, p_2, \dots, p_k$. Define

$$n = 1 + p_1 p_2 \cdots p_k.$$

Since any positive integer different from 1 is divisible by at least one prime number, n must be divisible by some p_i , where $1 \leq i \leq k$. This is a contradiction because n leaves a remainder of 1 when divided by any p_i . Thus, n cannot be divisible by any of the primes $p_1, p_2, \dots, p_k$. Consequently, there must be infinitely many prime numbers.

1.4 Proof by induction

To prove that $\forall n \in \mathbb{N}, n \geq n_0$ implies $P(n)$ is true, where $P(n)$ is a property depending on the nonnegative integer n , we use the principle of mathematical induction.

The proof by induction consists of two main steps:

1. Base Case: Show that $P(n_0)$ is true for some $n_0 \in \mathbb{N}$. This establishes that the property $P(n)$ holds for an initial value n_0 .
2. Inductive Step: Assume that $P(n)$ is true for some arbitrary integer $n \geq n_0$. This is known as the inductive hypothesis. Then, under this assumption, prove that $P(n + 1)$ is also true.

By demonstrating these two steps, we show that if $P(n_0)$ holds, and if $P(n)$ being true implies $P(n + 1)$ is true, then $P(n)$ holds for all $n \geq n_0$. This completes the proof by induction.

Example 4

1. Let us show that for any natural number n , $n^3 - n$ is a multiple of 3. We will use mathematical induction to prove this. For $n = 0$:

$$0^3 - 0 = 0,$$

which is divisible by 3. Thus, $P(0)$ is valid. Assume that $P(k)$ is valid, that is, $k^3 - k$ is a multiple of 3. We need to prove that $P(k+1)$ is valid, which means we need to show that $(k+1)^3 - (k+1)$ is a multiple of 3. Compute $(k+1)^3 - (k+1)$:

$$\begin{aligned}
(k+1)^3 - (k+1) &= (k+1) [(k+1)^2 - 1] \\
&= (k+1) [k^2 + 2k + 1 - 1] \\
&= (k+1)(k^2 + 2k) \\
&= (k+1)k^2 + 2(k+1)k \\
&= k^3 + 3k^2 + 3k + 1 - k - 1 \\
&= k^3 + 3k^2 + 2k.
\end{aligned}$$

Now, by the induction hypothesis, $k^3 - k = 3q$ for some $q \in \mathbb{N}$. Thus:

$$\begin{aligned}
(k+1)^3 - (k+1) &= k^3 - k + 3k^2 + 3k \\
&= 3q + 3(k^2 + k) \\
&= 3(q + k^2 + k).
\end{aligned}$$

Since $3(q + k^2 + k)$ is clearly a multiple of 3, $(k+1)^3 - (k+1)$ is also a multiple of 3. Thus, by induction, $n^3 - n$ is a multiple of 3 for all natural numbers n .

2. Let us show that for any $n \in \mathbb{N}^*$ and for any positive real x , $(1+x)^n \geq 1 + nx$.

We use mathematical induction to prove this.

For $n = 1$ we have

$$(1+x)^1 = 1+x.$$

Thus, $(1+x)^1 \geq 1 + 1 \cdot x$, which is true. Therefore, $P(1)$ is valid.

Assume that for some $k \in \mathbb{N}^*$, the property $P(k)$ is valid, i.e.,

$$(1+x)^k \geq 1 + kx$$

We need to show that $P(k+1)$ is also valid, i.e.,

$$(1+x)^{k+1} \geq 1 + (k+1)x.$$

To prove this, we have:

$$(1+x)^{k+1} = (1+x)(1+x)^k.$$

By the induction hypothesis, we know that

$$(1+x)^k \geq 1 + kx.$$

Thus,

$$(1+x)^{k+1} \geq (1+x)(1+kx).$$

Expanding this product:

$$\begin{aligned}(1+x)(1+kx) &= (1+x) + (1+x)kx \\ &= 1+x+kx+kx^2 \\ &= 1+(k+1)x+kx^2.\end{aligned}$$

Since $kx^2 \geq 0$ for any $x \geq 0$ and $k \geq 0$, we have:

$$1+(k+1)x+kx^2 \geq 1+(k+1)x.$$

Then $(1+x)^{k+1} \geq 1+(k+1)x$.

1.5 Proof by counter-example

Proof by counter-example is a method used to disprove an assertion by providing a single example that contradicts it. While an example cannot prove an assertion to be true, a counter-example is sufficient to prove that an assertion is false. To disprove a statement, it is often enough to find just one particular case that does not satisfy the assertion. This specific case is referred to as a counter-example.

In other words, if we can find an instance where the assertion fails, then the assertion is proven false. This is because the existence of one counter-example invalidates the claim made by the assertion.

Example 5

1. Assertion: If x and y are two real numbers such that $x^2 < y^2$, then $x < y$. This assertion is false.

Counter-Example: Consider $x = 2$ and $y = -3$. We have:

$$2^2 = 4 \quad \text{and} \quad (-3)^2 = 9,$$

so $2^2 < (-3)^2$ (i.e., $4 < 9$). However, 2 is not less than -3 . Thus, this counter-example shows that $x < y$ does not necessarily follow from $x^2 < y^2$.

2. Assertion: If $2^n - 1$ is prime, then n is prime. The converse of this statement is false.

Counter-Examples: Consider $n = 11$ and $n = 23$, both of which are prime numbers. We have:

$$2^{11} - 1 = 2047 = 23 \times 89,$$

and

$$2^{23} - 1 = 8388607 = 47 \times 178481.$$

In both cases, $2^n - 1$ is not a prime number despite n being prime. These counter-examples show that n being prime does not guarantee that $2^n - 1$ is prime.

2 Maps

Let E and F be two sets. A map f from E into F is defined as a triplet (E, F, G_f) , where G_f is a subset of $E \times F$ satisfying the following property:

$$\forall x \in E, \exists! y \in F : (x, y) \in G_f.$$

Intuitively, a map from E into F is a correspondence (or process) that associates each element of E with a unique element y in F . Here:

- The set E is called the domain (or source) of f .
- The set F is called the codomain (or target) of f .
- The element y is called the image of x under f and is expressed as $y = f(x)$.
- The element x is called the preimage or antecedent of y under f .
- We can represent the map f using the notation:

$$\begin{array}{ccc} f : & E & \longrightarrow & F \\ & x & \longmapsto & f(x). \end{array}$$

- This notation indicates that f maps elements from the set E to elements in the set F , and for each $x \in E$, the image $f(x)$ is uniquely determined in F .
- We denote by E^F the set of all maps from E into F .

Example 6

1. Let $E = \{a, b, c\}$, $F = \{1, 2, 3\}$, and let G_f, G_g be the two parts of $E \times F$ given by

$$G_f = \{(a, 1), (b, 1), (a, 2), (c, 3)\}$$

and

$$G_g = \{(a, 2), (b, 1), (c, 3)\}.$$

It is clear that f is not a map but g is a map from E into F .

2. For any set E the following map

$$\begin{array}{ccc} id_E : & E & \longrightarrow & E \\ & x & \longmapsto & x \end{array}$$

is called identity map of E .

3. Let E and F be two sets. For a fixed element a in F . The map

$$\begin{array}{ccc} f_a : & E & \longrightarrow & F \\ & x & \longmapsto & a \end{array}$$

is called constant. This map sends every element of E to the same element a in F .

4. Let E and E' be two sets such that $E' \subset E$. The map

$$\begin{array}{ccc} i_A : E' & \longrightarrow & E \\ x & \longmapsto & x \end{array}$$

is called the canonical injection. This map embeds E' into E by mapping each element of E' to itself in E .

5. Let E and E' be two sets such that $E' \subset E$. The map

$$\begin{array}{ccc} \chi_{E'} : E & \longrightarrow & \{0, 1\} \\ x & \longmapsto & \begin{cases} 1 & \text{if } x \in E' \\ 0 & \text{otherwise} \end{cases} \end{array}$$

is called the indicator map of E' . This map indicates membership in E' by mapping elements to 1 if they are in E' and to 0 otherwise.

6. Let E and F be two sets. The map

$$\begin{array}{ccc} \rho_E : E \times F & \longrightarrow & E \\ (x, y) & \longmapsto & x \end{array}$$

is called the projection map over E . In the same way the map

$$\begin{array}{ccc} \rho_F : E \times F & \longrightarrow & F \\ (x, y) & \longmapsto & y \end{array}$$

is called the projection map over F . These maps extract the first and second components, respectively, from the pairs in $E \times F$.

7. Let f_1 and f_2 be two maps defined as follows

$$\begin{array}{ccc} f_1 : E_1 & \longrightarrow & F_1 \\ x & \longmapsto & f_1(x) \end{array}$$

and

$$\begin{array}{ccc} f_2 : E_2 & \longrightarrow & F_2 \\ y & \longmapsto & f_2(y). \end{array}$$

The map product is defined as

$$\begin{array}{ccc} f_1 \times f_2 : E_1 \times E_2 & \longrightarrow & F_1 \times F_2 \\ (x, y) & \longmapsto & (f_1(x), f_2(y)). \end{array}$$

8. A choice function on a set S is any map f from $\mathcal{P}(S)$ into S such that for any $T \subset S$ we have $f(T) \in T$.

Consider the set $\mathbb{N}$. It is known that every subset has a smallest element. We can therefore define the function $F : \mathcal{P}(\mathbb{N}) \rightarrow \mathbb{N}$, which transforms any subset of $\mathbb{N}$ into its smallest element. This function satisfies: for every $X \in \mathcal{P}(\mathbb{N})$, we have $F(X) \in X$; in other words, it allows us to "choose" an element from each subset of the set $\mathbb{N}$. We say that F is a choice function on $\mathbb{N}$. Note that for a set like $\mathbb{Q}$, we do not explicitly know such a function, but we can demonstrate its existence using the axiom of choice.

Recall that one of the equivalent formulations of the choice axiom is:

For any collection of non-empty, pairwise disjoint subsets of a set A , there exists a subset of A that intersects each of these subsets in exactly one element.

Two maps

$$\begin{array}{ccc} f : E & \longrightarrow & F \\ x & \longmapsto & f(x), \end{array}$$

and

$$\begin{array}{ccc} g : E' & \longrightarrow & F' \\ x & \longmapsto & g(x) \end{array}$$

are said to be equal if $E = E'$, $F = F'$, and $\forall x \in E: f(x) = g(x)$.

Example 7 *The following three maps are not equal*

$$\begin{array}{ccc} f : [0, +\infty[& \longrightarrow & \mathbb{R} \\ x & \longmapsto & x^2, \end{array}$$

$$\begin{array}{ccc} g : [0, +\infty[& \longrightarrow & [0, +\infty[\\ x & \longmapsto & x^2, \end{array}$$

$$\begin{array}{ccc} h : \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longmapsto & x^2, \end{array}$$

because, even the three formulas are equal, the sources and the targets are not equal. But there are relations between them, the following notions clarify this.